


```

C:\AyushSR\crypto\lab16.exe X +
Enter two primes p and q : 3 11

---RSA Key Generation---
Public key = (3, 33)
Private key = (7, 33)
Enter message : HELLO
Encrypted : 13 31 11 11 5
Decrypted : HELLO

```

```

C:\AyushSR\crypto\lab17.exe X +
Enter prime p and primitive root g : 23 5
Enter private key x : 6
Enter random key k : 3
Enter message m (m < p) : 10

---ElGamal Cryptographic System---
Public key (p,g,y) = (23,5,8)
Ciphertext = (10,14)
Decrypted message = 10

```

```

C:\AyushSR\crypto\lab18.exe X +
Enter number of sandbox file names : 3
Enter file names :
file.exe note.txt run.bat

Scan Report :
file.exe -> suspicious
note.txt -> normal
run.bat -> suspicious

Enter agent name : agentA
Enter number of copies : 2

Replication Log :
copy_1_agentA
copy_2_agentA

Simulation only.

```

```

C:\ayush\crypto\lab1.exe X
Enter text: iamayush
Enter key (0-25): 3
Cipher: LDPDBXVK
Plain : IAMAYUSH

```

```

C:\ayush\crypto\lab2.exe X
1. Encrypt
2. Decrypt
Choose: 2
Enter key: KEY
Enter text: QIKNSA
Plaintext (raw): PLAINX

```

```

C:\ayush\crypto\lab2.exe X
1. Encrypt
2. Decrypt
Choose: 1
Enter key: KEY
Enter text: PLAIN
Ciphertext: QIKNSA

```

```

C:\ayush\crypto\lab5.exe X +
One-Time Pad (OTP)
1) Encrypt
2) Decrypt
Choose: 1
Enter text: HELLO
Enter key (same length as text): XMCKL

Ciphertext: EQNVZ

```

```

C:\ayush\crypto\lab3.exe X +
Enter 2x2 key matrix (a b c d): 3 3 2 5

1) Encrypt
2) Decrypt
Choose: 2
Enter text: HIOZHN

Plaintext: HELLOX

```

```

C:\ayush\crypto\lab4.exe X
Vigenere Cipher
1) Encrypt
2) Decrypt
Choose: 1
Enter text: ATTACK AT DAWN
Enter key: LEMON

Ciphertext: LXFOPVEFRNHR

```

```

C:\ayush\crypto\lab3.exe X +
Enter 2x2 key matrix (a b c d): 3 3 2 5

1) Encrypt
2) Decrypt
Choose: 1
Enter text: HELLO

Ciphertext: HIOZHN

```

```
C:\ayush\crypto\lab6.exe X + v
Vernam Cipher (Binary XOR)
1) Encrypt
2) Decrypt
Choose: 1
Enter plaintext: HELLO
Enter key (same length): XMCKL

Step-by-step XOR (8-bit):
P_char P_bin      K_char K_bin      C_bin
H      01001000  X      01011000  00010000
E      01000101  M      01001101  00001000
L      01001100  C      01000011  00001111
L      01001100  K      01001011  00000111
O      01001111  L      01001100  00000011

Ciphertext (numbers): 16 8 15 7 3
(Use these numbers for Decrypt option)
```

```
C:\ayush\crypto\lab5.exe X + v
One-Time Pad (OTP)
1) Encrypt
2) Decrypt
Choose: 2
Enter text: EQNVZ
Enter key (same length as text): XMCKL

Plaintext: HELLO
```

```
C:\ayush\crypto\lab6.exe X + v
Vernam Cipher (Binary XOR)
1) Encrypt
2) Decrypt
Choose: 2
Enter ciphertext numbers (space separated): 16 8 15 7 3
Enter key (same length as number count): XMCKL

Step-by-step XOR (8-bit):
C_num C_bin      K_char K_bin      P_bin
16     00010000  X      01011000  01001000
8      00001000  M      01001101  01000101
15     00001111  C      01000011  01001100
7      00000111  K      01001011  01001100
3      00000011  L      01001100  01001111

Plaintext: HELLO
```